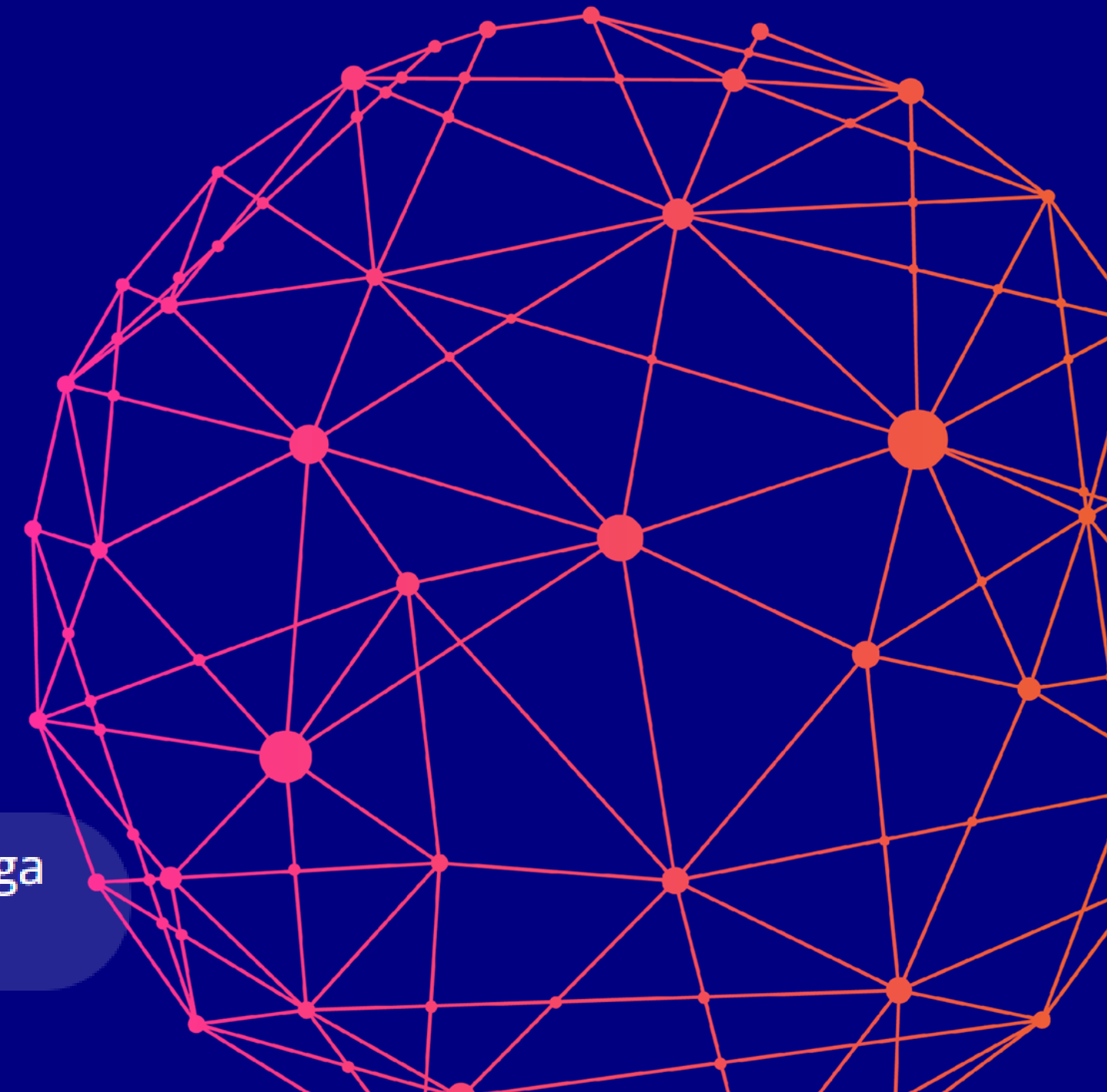


Data Spaces Symposium

Accelerating adoption. Increasing impact.

Sovereign B2G Data Spaces:

Secure and Interoperable Urban
Infrastructure Data Sharing
with Eclipse Components



Lucía Alcibar-Arechuluaga
Inkolan



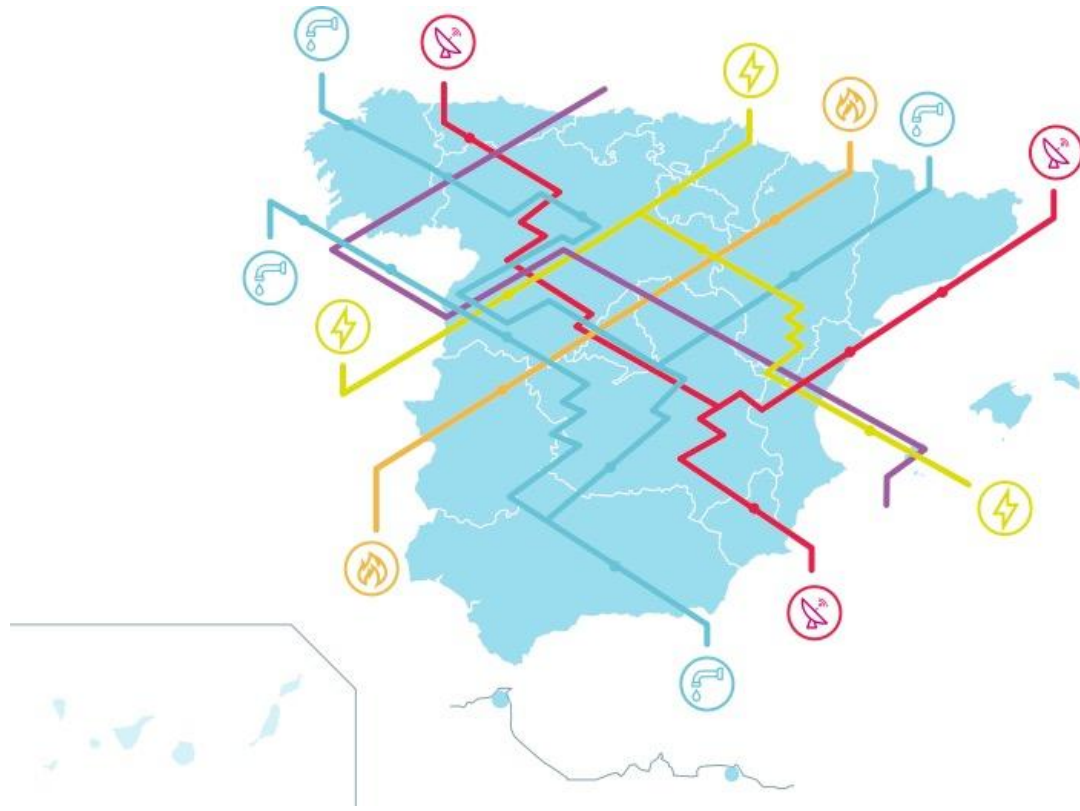
Enrique Areizaga
Tecnalia

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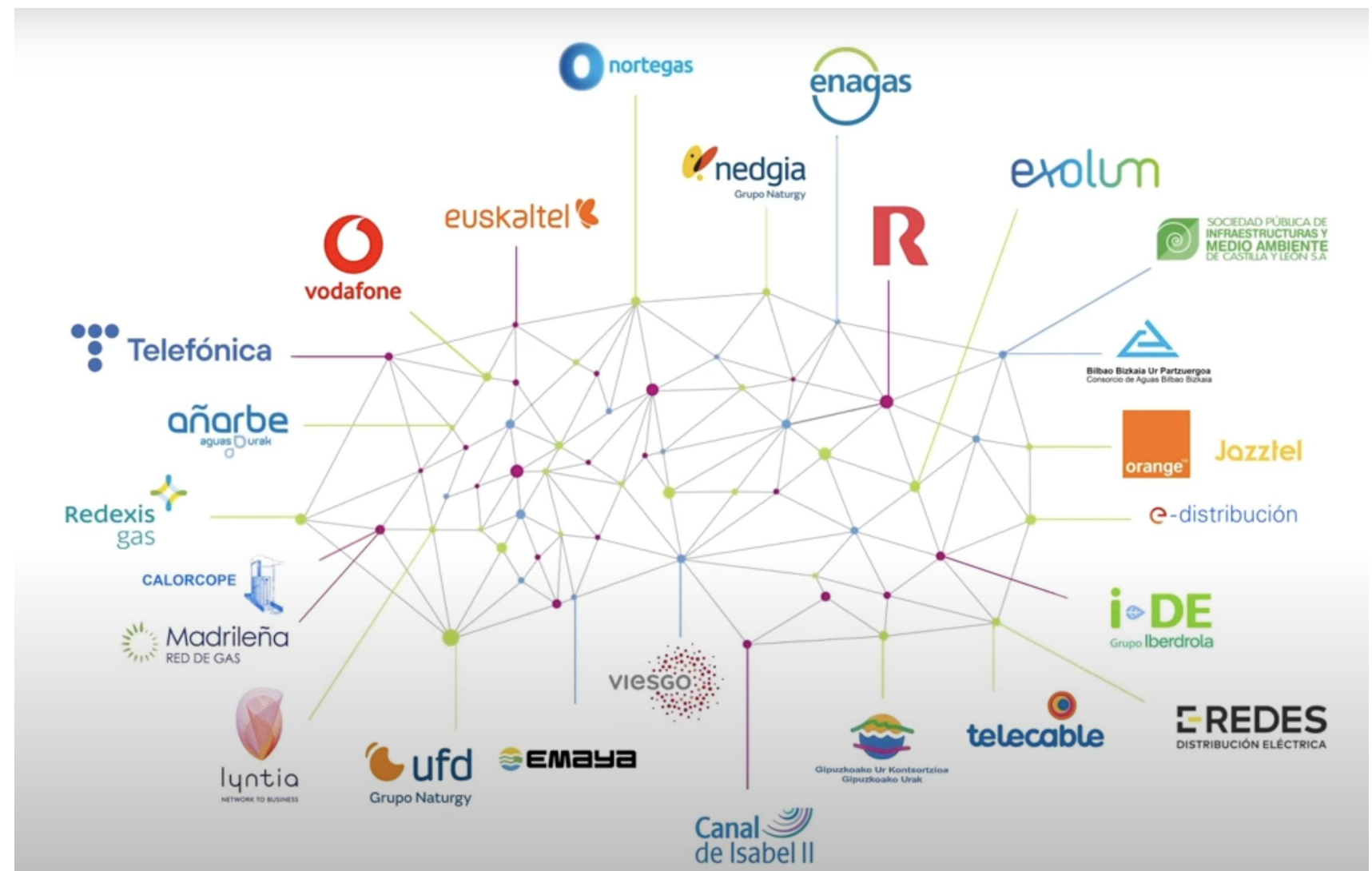
A non-profit group made up of most of the country's major utility operators.

They provide us with all the information so that Inkolan can facilitate access to it in a unified, simple and digital way



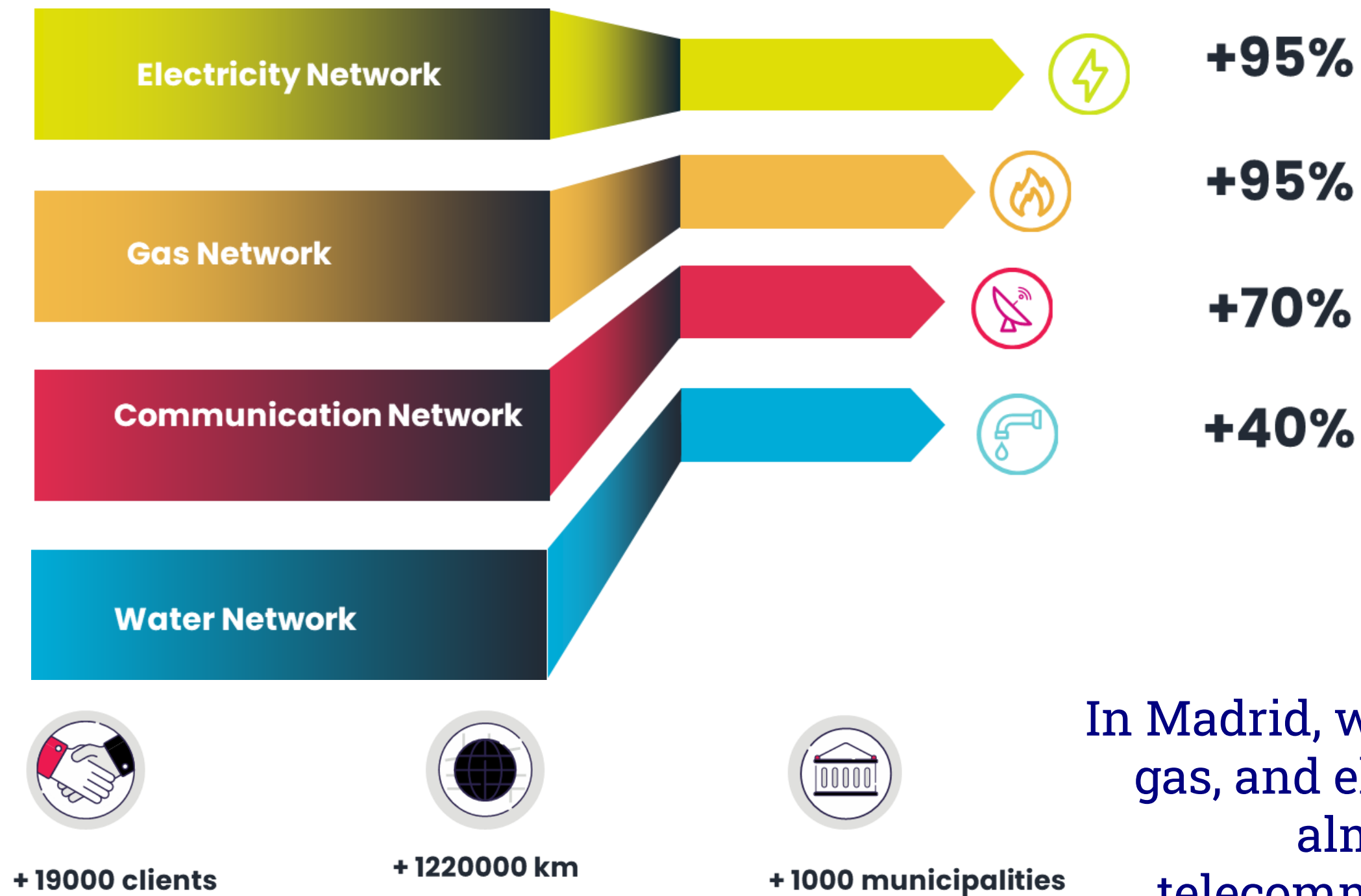
Our data backs us up. We continue to grow every year to provide with the best service.

Proof of this is the constant increase in requests for downloading our files, our growing presence in City Councils, Institutions and Large Companies, as well as a network information coverage of service networks in the territory of more than 1.2MM Km



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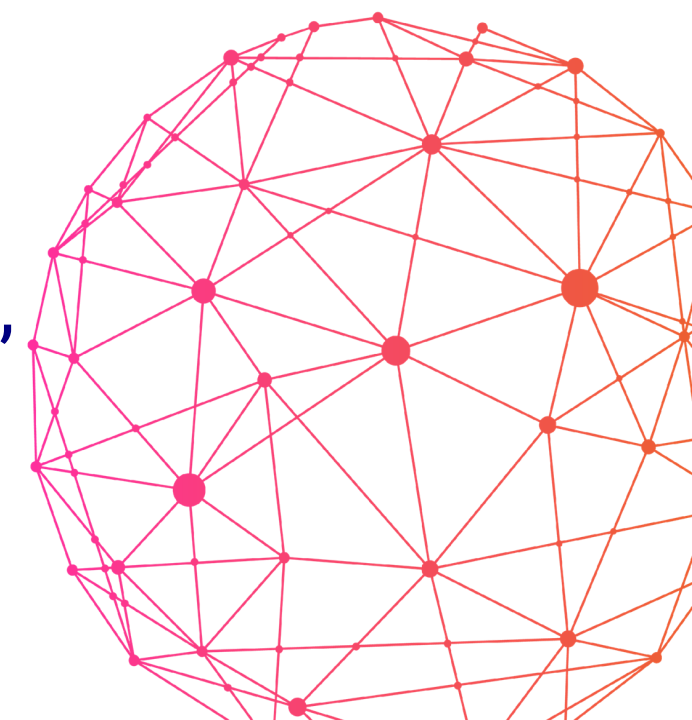
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We work hard to increase the number of companies and public services everyday.

Every year a new company starts to work with us

In Madrid, we have 100% of the water, gas, and electricity networks and almost 90% of their telecommunications networks.



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300 Clients download
information every
day



1.056.489 Km
From telecom and utility
networks stored in the system



INSPIRE

Network based on the
standard for service networks



OGC

Allows data access through
standards (WMS / WFS / WCS)



INTEROPERABILITY

Access via OGC and REST
services



MULTIPLE FORMATS

Supports multiple (CAD / GIS)
formats for data input and
output



ETRS89 SYSTEM

Data normalized to the
ETRS89 coordinate system.



REPROJECTION

Allows importing or exporting
information to other coordinate
systems



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Transforming Inkolan into a Secure, Scalable Data Space

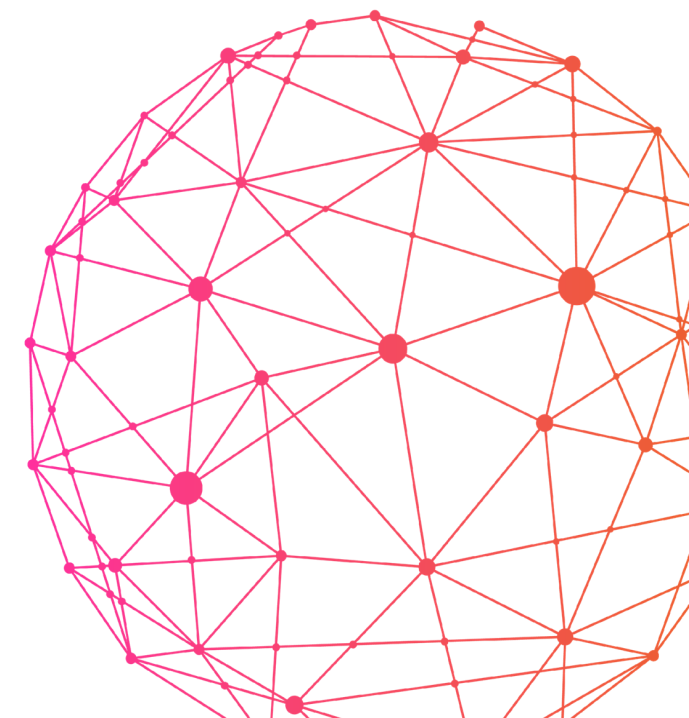
Inkolan already had all the pieces

- Data Providers: Energy, gas, water, telecom companies
- Data Consumers: Public admin, construction, individuals
- Marketplace: Select, pay, download

Inkolan wasn't starting from scratch. We already had the core components: providers, consumers, and a marketplace.

The challenge?

Making it a trusted, secure, and interoperable Data Space.



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The Critical Challenge – Data Accuracy & Trust

Why Data Quality is Non-Negotiable

Errors = Real time consequences

- Example: Pipeline damage → public safety risk
- Legal & financial liability

Need for real-time, reliable data

Heterogeneous data sources → inconsistency

"Imagine a construction team digging based on outdated utility data. One mistake could shut down a city. That's why accuracy isn't just a feature—it's a responsibility."



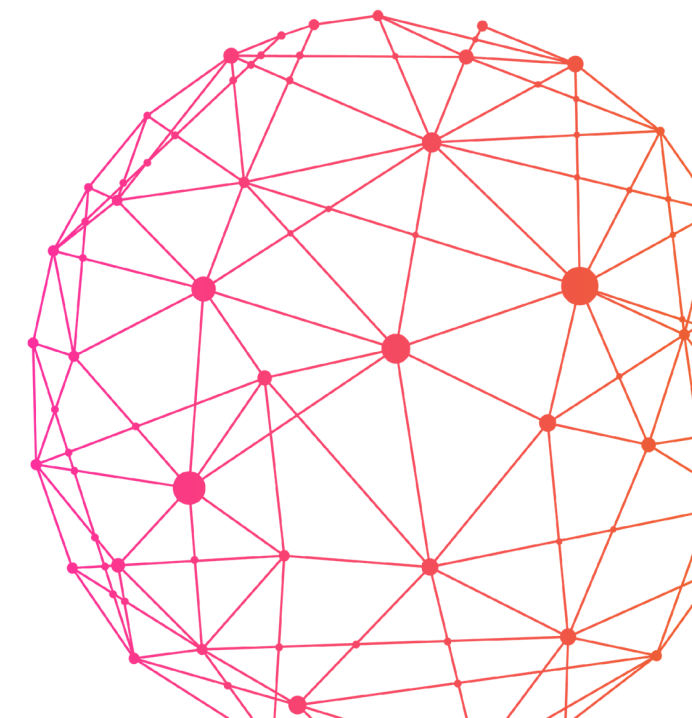
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Building Trust with Standards & Governance

Standardizing for Trust and Accountability

- Standardize data uploads from providers
- Enforce access & usage policies → digital contracts
- Built on **international standards**:
 - ISO/IEC 20151 (Data Space Concepts)
 - ISO/IEC DIS 26450 (DSP)
 - ISO/IEC DIS 26451 (DCP)
- Full process tracking for audits

"We didn't reinvent the wheel. We built on proven international standards—so every data exchange is secure, auditable, and legally defensible."



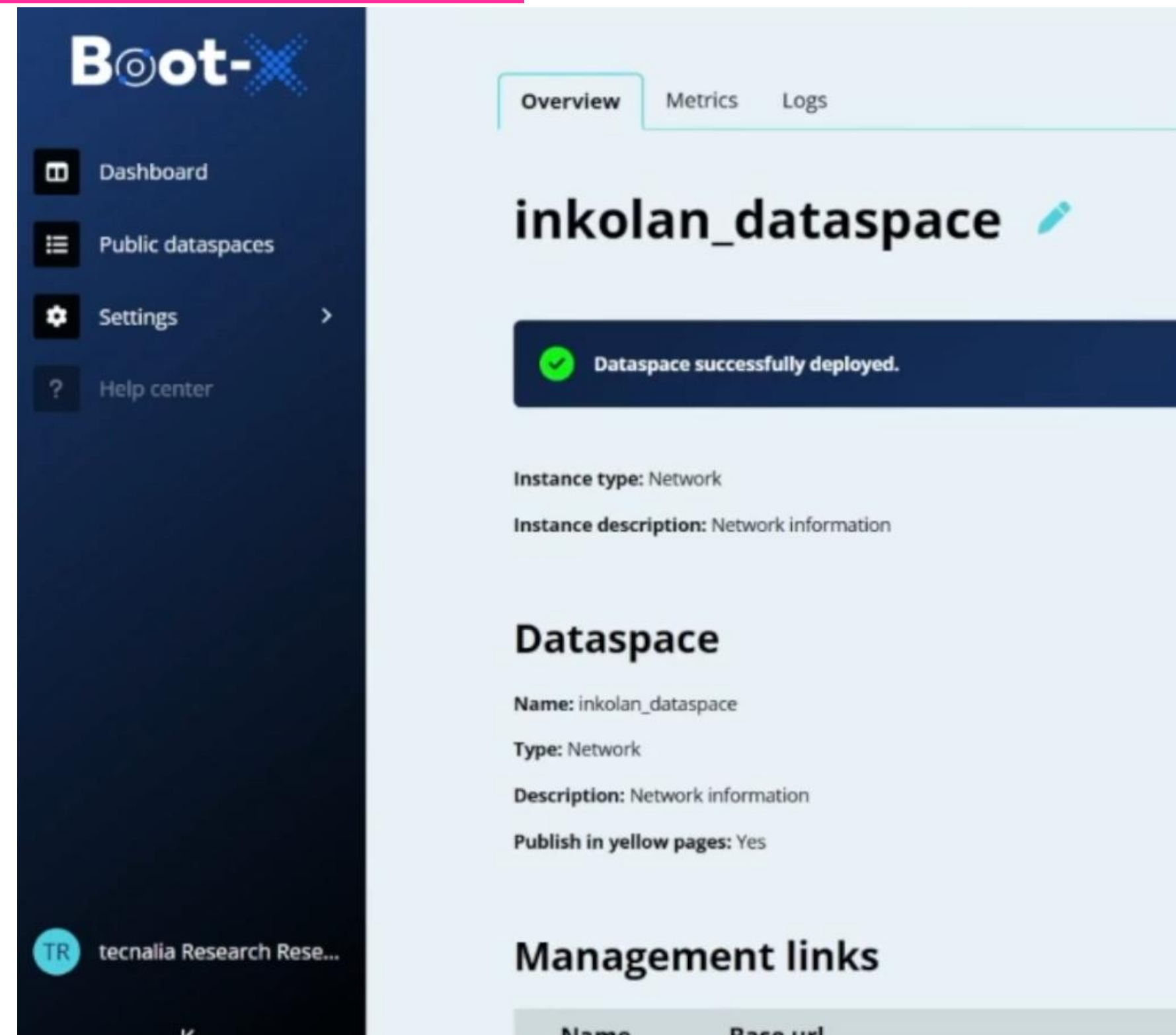
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Accelerating Migration with Gaia-X & Open Source

Fast-Tracking the Transition

- Partnered with Boot-X, a Gaia-X lighthouse & IDSA-certified solution
- Built on Eclipse Dataspace Components (EDC)
- Open-source, reliable, cloud-native

"We leveraged a trusted, certified partner—Boot-X—that brought ready-to-use, open-source components to speed up our journey."



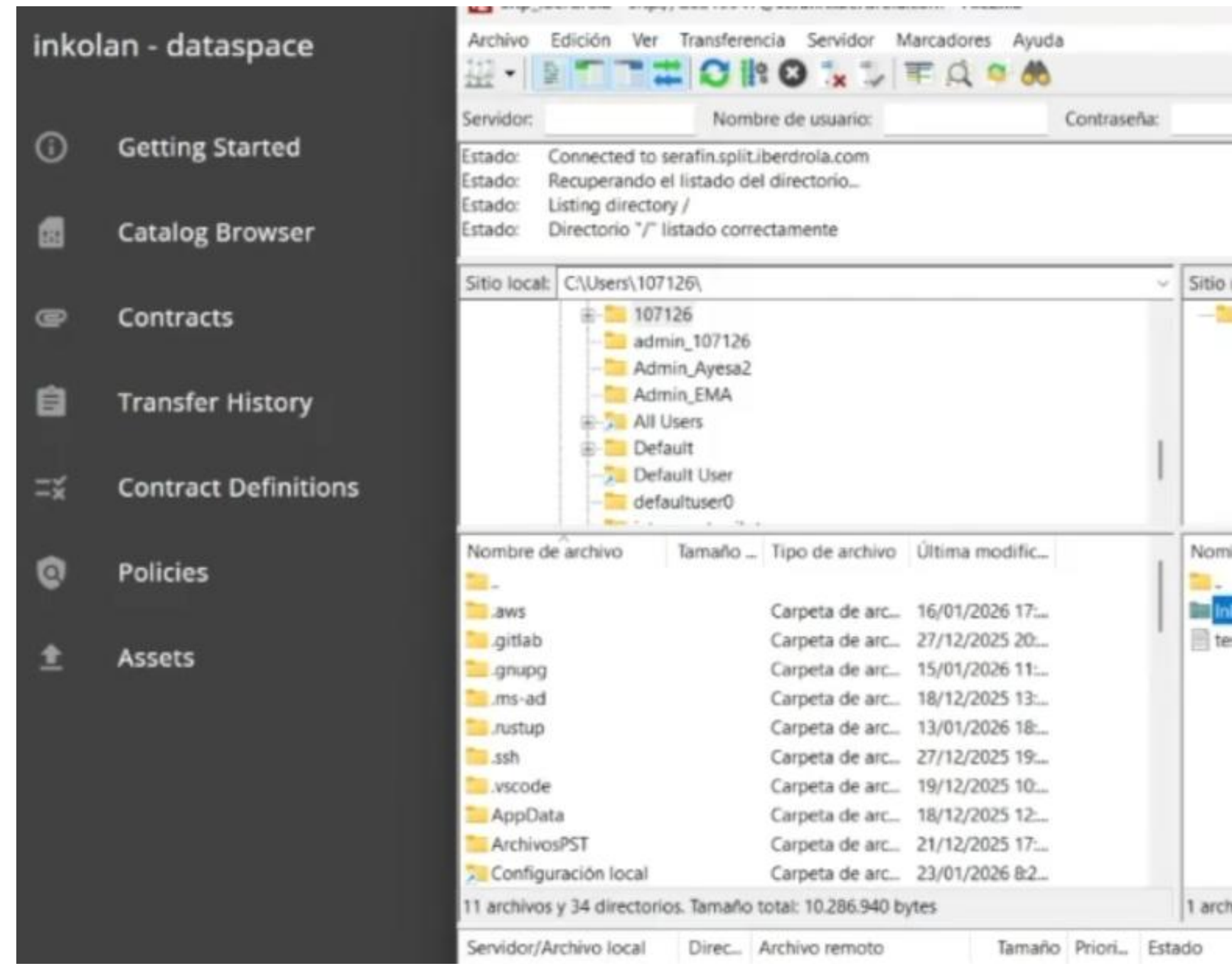
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Overcoming Adoption Barriers

Making Change Easy for Providers

- Providers used **SFTP**—new EDC approach = unfamiliar
- Uncertainty → **delays in migration**
- **Solution:** Collaborated with EDC team to build an SFTP extension
- **Result:** Suppliers kept their familiar workflow

"Change is hard. But by adapting the technology to their existing tools—instead of forcing them to change—we made adoption smooth and fast."



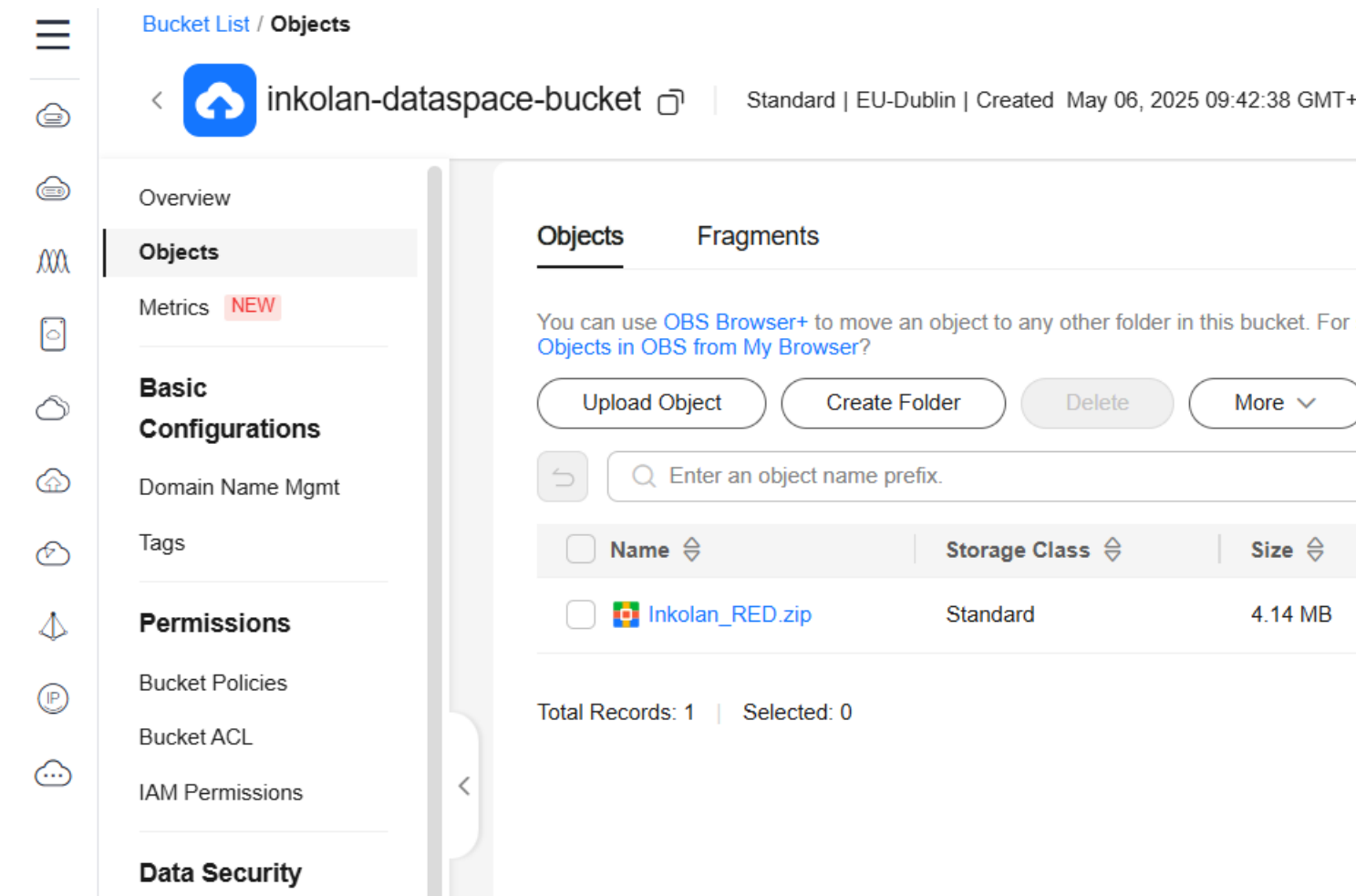
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Securing Data Sovereignty & Governance

Data Sovereignty in the Cloud

- Secure cloud storage systems
- Reinforce trust, control, and compliance
- **Data stays under control** of the provider
- **No unauthorized access or leaks**

"We didn't just move data to the cloud—we built a secure, governed environment where data sovereignty is preserved, even in a shared space."



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Solving the Consumer Challenge

**Safety Access for Public Administration – Viewing,
Not Storing**

Public admin: free access, but data must be traceable

Problem: Download links → outdated information

Solution:

- Access via consumer EDC connectors
- Data viewed, not stored
- Policies applied in real-time

"Public admins need data for planning—
but we needed to protect this data. So, we
made it viewable and updated. That way,
data is accessible but secure.

Traceability is essential for Utilities"



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Observability – The Audit Trail

Transparency by Design

Observability built into the system

Tracks:

- Incidents
- Availability
- Security events

Irreversible logs for audits

Respects data sovereignty

"Every action is logged—so if something goes wrong, we know exactly who, when, and why. That's accountability in action"



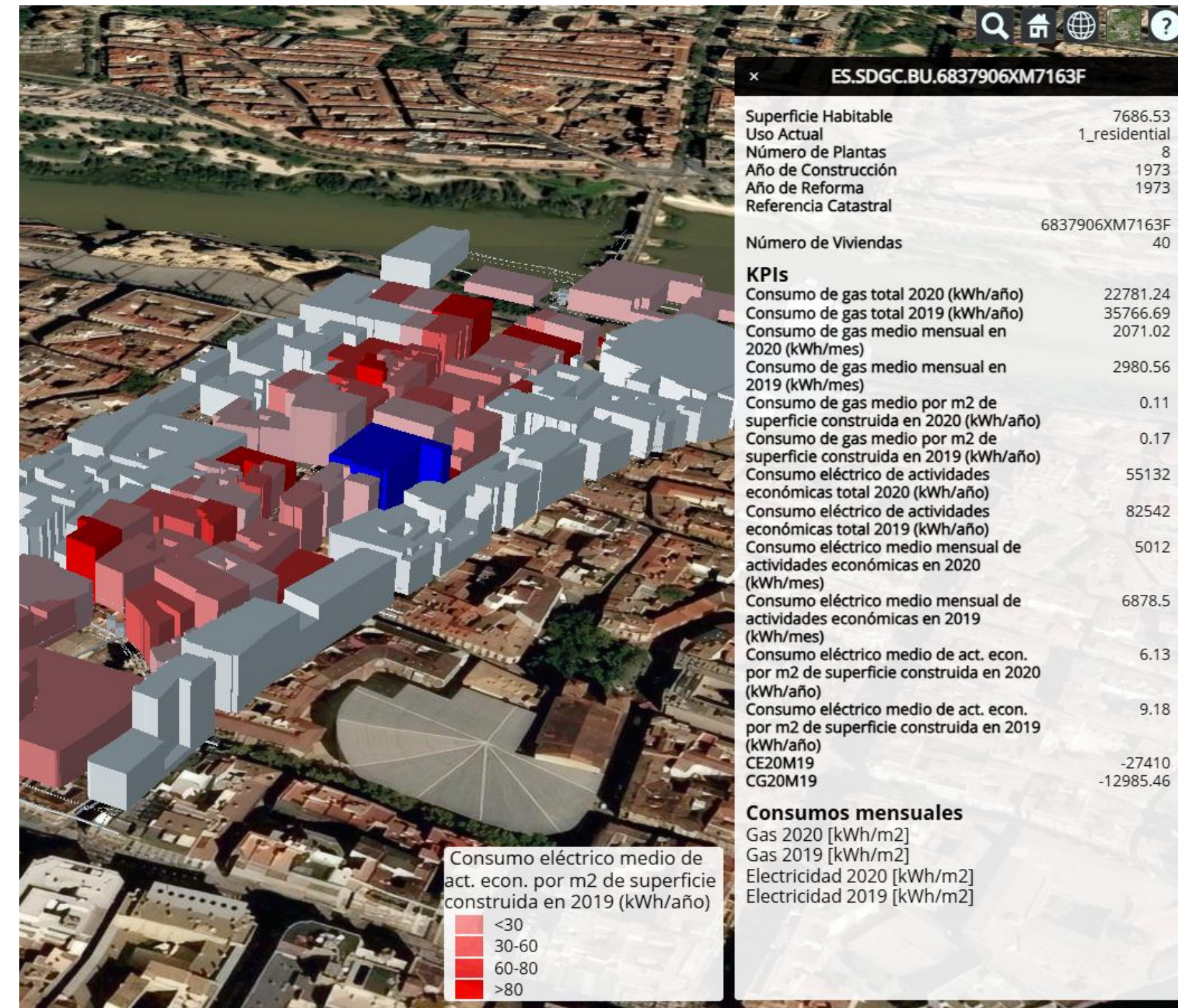
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Scalable & Cost-Effective Architecture

Low-Cost. High Scalability Model

- **Providers:** Join & assign services
- **Consumers:** Get “leased” connectors -> Go into “sleep mode” after use
- **Zero cost after consumption**
- **Future-proof:** Bring your own connectors

"We made it affordable. Consumers pay only for what they use—no ongoing costs."



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Key Learnings from the Pilot

Lessons Learned: What Makes a Data Space Work?

Interoperable -> Use international standards

Support the current process -> integrate existing technologies

Easy to participate -> leverage existing dataspace services

Cheap and flexible -> leverage cloud-native capabilities



The Bigger Picture – A Trend Toward Open, Context-Independent Spaces

The trend towards context-independent dataspace solutions can be observed in other initiatives as well. Current open source implementations around the EDC provide the required flexibility for such service offerings by (European) Cloud Service Provider, while lowering the costs for everyone.

"We'd love to share more. Come to our session. Let's build the future of data together."

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Accelerating adoption. Increasing impact.

Thank you for your kind attention.

Inkolan – Where Data Flows
Securely, Responsibly, and
Openly

inkolan

tecnal:a
MEMBER OF BASQUE RESEARCH
& TECHNOLOGY ALLIANCE

